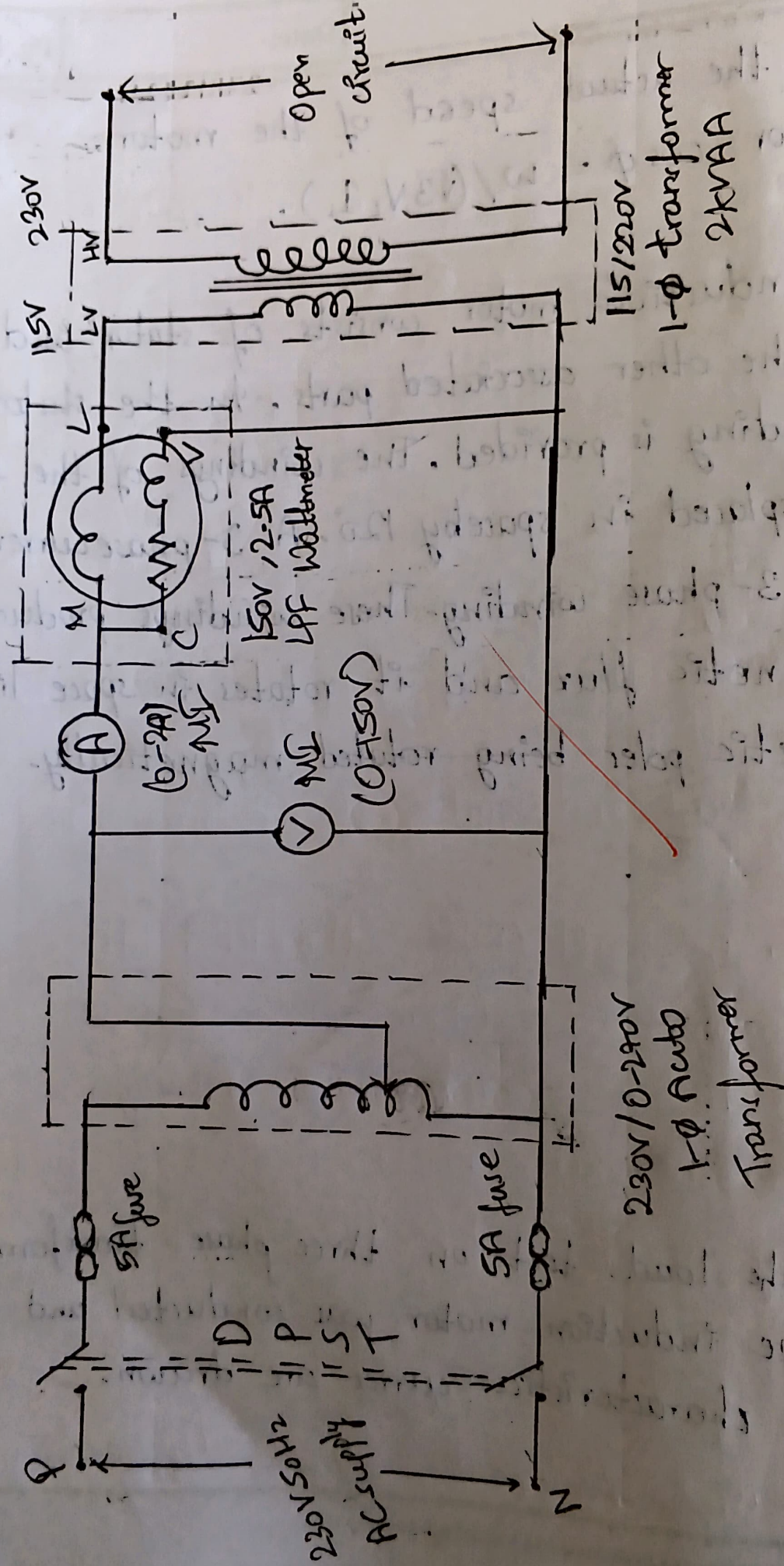


OPEN CIRCUIT TEST:



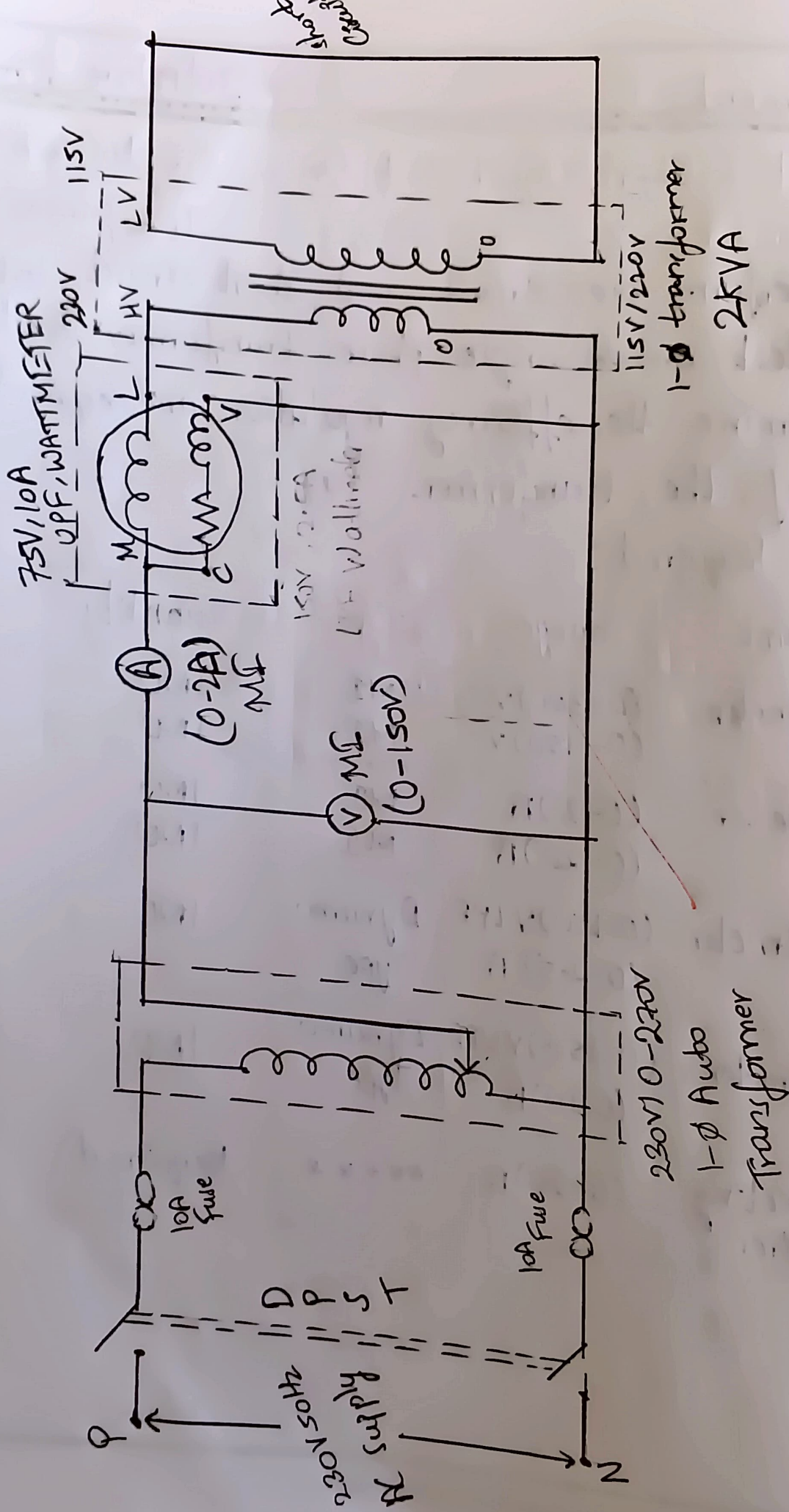
Open Circuit & Short Circuit Test On a Single ~~Test~~ Phase Transformer

Aim: To perform open circuit and short circuit test on a single test on a single phase transformer and to pre-determine the efficiency, regulation and equivalent circuit of the transformer.

Apparatus Required:

S.NO	Name	Range	Type	Quantity
1	Voltmeter	(0-300)V	MI	1 NO
		(0-150)V	MI	1 NO
2	Ammeter	(0-2)A	MI	1 NO
		(0-20)A	MI	1 NO
3	Wattmeter	(0-150)V LPF (0-2.5) A	Dynamo Type	1 NO
4	Wattmeter	(0-150)V UPF (0-10)A	Dynamo Type	1 NO
5	Connecting Wires	(0-20)A	*****	Required

SHORT CIRCUIT TEST:



Transformer Specifications:

Transformer Rating : 500 VA

Winding Details:

LV (in volts) : 115V

LV side current : 4.34A

HV (in volts) : 230V

HV side current : 2.17A

Type (Shell/core) : Core

Auto transformer Specifications:

Input voltage (in volts) : 115V

Output voltage (in volts) : 230V

Frequency (in Hz) : 50Hz

Current rating (in Amp) : 10A

Procedure:

Open circuit test:

1. Connections are made as per the circuit diagram.
2. Ensure that variac is set to zero output voltage position before starting the experiment.
3. Switch ON the supply. Now supply the rated voltage to the primary winding by using variac.



4. The readings of the voltmeter, ammeter and wattmeter are noted down in tabular form.
5. Then variac is set to zero output position and switch OFF the supply.
6. Calculate R_0 and X_0 from the readings.

Short Circuit Test:

1. Connections are made as per the circuit diagram.
2. Ensure that variac is set to zero output voltage position before starting the experiment.
3. Switch ON the supply. Now apply the rated current to the Primary winding by using Variac.
4. The readings of the voltmeter, ammeter and wattmeter are noted down in Tabular form.
5. Then variac is set to zero output position and switch OFF the supply.
6. Calculate R_{01} and X_{01} from the readings.

Observations

I) For OC Test

S.NO	Voltmeter Reading (V_o) in volts	Ammeter Reading (I_o) in Amps	Wattmeter Reading (W_o) in Watts
1	115	0.246	20

II) For SC Test

S.NO	Voltmeter Reading (V_{sc}) in volts	Ammeter Reading (I_{sc}) in Amps	Wattmeter reading (W_{sc}) in Watts
1	6.5	2.2	30

calculation :-

for O.C Test :-

$$V_o = 115 \text{ V}, \quad I_o = 0.246, \quad P_o = 20 \text{ W}$$

$$P_o = V_o I_o \cos \phi_o$$

$$\cos \phi_o = \frac{P_o}{V_o I_o}$$

$$= \frac{20}{115 \times 0.246}$$

$$= 0.706 \text{ lag.}$$

$$I_w = I_o \cos \phi_o = 0.246 (0.706) = 0.173 \text{ A}$$

$$\sin \phi_0 = \sqrt{1 - \cos^2 \phi_0}$$

$$= 0.708$$

$$I_m = I_0 \sin \phi_0$$

$$= 0.246 \times 0.708$$

$$= 0.174 \text{ A}$$

$$R_{01} = \frac{V_0}{I_m} = \frac{115}{0.173} = 664.7 \Omega$$

$$X_0 = \frac{V_0}{I_m} = \frac{115}{0.173} = 664.7 \Omega$$

for s.c Test :

$$V_{sc} = 6.5 \text{ V}, \quad I_{sc} = 2.2 \text{ A}, \quad P_{sc} = 30 \text{ watts}$$

$$R_a = \frac{P_{sc}}{I_{sc}^2} \quad \left[\because P_{sc} = I_{sc}^2 R_{sc} \right]$$

$$= \frac{30}{(2.2)^2} = 6.198 \Omega$$

$$Z_a = \frac{V_{sc}}{I_{sc}} = \frac{6.5}{2.2} = 2.954 \Omega$$

$$X_a = \sqrt{Z_a^2 - R_a^2} = 5.450 \Omega$$

Result: Hence, conduction of OC & SC circuit test on a single phase transformer and calibration of its parameters.

At

S. G. 311